

Music VAE

Music VAE (variational auto encoder) is a generative model that can be used to create novel sequences or interpolate between existing sequences.

The model initialisation can be done with the following code snippet:

```
# Import dependencies.
from magenta.models.music_vae import configs
from magenta.models.music_vae.trained_model import TrainedModel

# Initialize the model.
print("Initializing Music VAE...")
music_vae = TrainedModel(
    configs.CONFIG_MAP['cat-mel_2bar_big'],
    batch_size=4,
    checkpoint_dir_or_path='/content/mel_2bar_big.ckpt')
```

New sequence creation may be carried out as shown below:

```
generated_sequences = music_vae.sample(n=2, length=80,
    temperature=1.0)

for ns in generated_sequences:
    mm.plot_sequence(ns)
    mm.play_sequence(ns, synth=mm.fluidsynth)
```

Magenta - JavaScript

As stated earlier, the JavaScript distribution of Magenta can be used right inside your Web browser. It provides many powerful features to manipulate music in the Web browser.

The Magenta music can be incorporated into JavaScript as follows:

```
<script src="https://cdn.jsdelivr.net/npm/@magenta/
music@^1.0.0"></script>
```

Managing NoteSequences and playing them is rather simple in JavaScript:

```
player = new mm.Player();
player.start(TWINKLE_TWINKLE);
player.stop();
```

The three machine learning models (MusicRNN, MusicVAE, and onsets and frames), which were explained earlier with Magenta – Python, are available in the JavaScript version as well. A sample code snippet for MusicRNN is shown below:

```
// Initialize the model.
music_rnn = new mm.MusicRNN('https://storage.googleapis.com/
magentadata/js/checkpoints/music_rnn/basic_rnn');
```

```
music_rnn.initialize();
```

```
// Create a player to play the sequence we'll get from the
model.
```

```
rnnPlayer = new mm.Player();
```

```
function play() {
    if (rnnPlayer.isPlaying()) {
        rnnPlayer.stop();
        return;
    }
```

```
    // The model expects a quantized sequence, and ours was
    unquantized:
```

```
    const qns = mm.sequences.quantizeNoteSequence(ORIGINAL_
    TWINKLE_TWINKLE, 4);
    music_rnn
        .continueSequence(qns, rnn_steps, rnn_temperature)
        .then((sample) => rnnPlayer.start(sample));
}
```

rnn_steps and rnn_temperature can be used to modify the output.

There are many demos available at <https://magenta.tensorflow.org/demos>.

The Magenta Studio provides a collection of music plugins that are built using Magenta models (<https://magenta.tensorflow.org/studio>). The power of Magenta can be understood from the variety of demos.

With the advent of advanced libraries such as Magenta, handling art with machine learning is becoming simpler and more effective. We can surely expect more to come in the near future.

To conclude, as the official documentation says, Magenta can really make your browser sing. **END** 🐧

References

- [1] Neural Style Transfer with TensorFlow: https://colab.research.google.com/github/tensorflow/docs/blob/master/site/en/tutorials/generative/style_transfer.ipynb
- [2] Magenta Music Python: https://colab.research.google.com/notebooks/magenta/hello_magenta/hello_magenta.ipynb#scrollTo=XKk8rGihUR6B
- [3] Magenta Music JavaScript: <https://hello-magenta.glitch.me/>
- [4] Magenta Demos: <https://magenta.tensorflow.org/demos>

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